

# LEVEL 1

## Description

Fix abstraction and decode transformed output.

## Code with Blanks

```
abstract class Secret {
    _____ void unlock();
}
class Vault extends Secret {
    String clue = "I have many teeth but cannot bite";
    @_____
    public _____ unlock() {
        char[] arr = clue.toCharArray();
        for(int i=0;i<arr.length/2;i++){
            char t=arr[i];
            arr[i]=arr[arr.length-1-i];
            arr[arr.length-1-i]=t;
        }
        String mid=new String(arr);
        String out="";
        for(int i=0;i<mid.length();i++){
            if(i%2==____){
                out+=mid.charAt(i);
            }
        }
        System.out.println("LEVEL 1 UNLOCKED");
        System.out.println(out);
    }
}
public class Main{
    public static void main(String[] args){
        Secret s=new Vault();
        s._____( );
    }
}
```

## Filled Code

```
abstract void unlock();
@Override
public void unlock(){
    char[] arr = clue.toCharArray();
    for(int i=0;i<arr.length/2;i++){
        char t=arr[i];
        arr[i]=arr[arr.length-1-i];
        arr[arr.length-1-i]=t;
    }
    String mid=new String(arr);
    String out="";
    for(int i=0;i<mid.length();i++){
        if(i%2==0){
            out+=mid.charAt(i);
        }
    }
    System.out.println("LEVEL 1 UNLOCKED");
    System.out.println(out);
}
s.unlock();
```

## Fill Answers

abstract, Override, void, 0, unlock

## Expected Output

LEVEL 1 UNLOCKED  
eibnnbhttaaaaaI

# LEVEL 2

## Description

Encapsulation + transformations.

## Code with Blanks

```
class DataBox{
    private String secret;
    public void _____(String input){
        String f="";
        for(int i=0;i<input.length();i++){
            char ch=input.charAt(i);
            if(Character._____(ch)){
                f+=ch;
            }
        }
        char[] arr=f.toCharArray();
        char last=arr[arr.length-1];
        for(int i=arr.length-1;i>0;i--){
            arr[i]=arr[i-1];
        }
        arr[0]=last;
        secret=new String(arr);
    }
    public String _____(){
        return secret;
    }
}

public class Main{
    public static void main(String[] args){
        DataBox obj=new DataBox();
        String encoded="P1I2E3";
        obj._____(encoded);
        String res=obj._____;
        char[] arr=res.toCharArray();
        for(int i=0;i<arr.length/2;i++){
            char t=arr[i];
            arr[i]=arr[arr.length-1-i];
            arr[arr.length-1-i]=t;
        }
        System.out.println("LEVEL 2 UNLOCKED");
        System.out.println(new String(arr));
    }
}
```

## Filled Code

```
setSecret, isLetter, getSecret
(setSecret + getSecret used correctly)
```

## Fill Answers

```
setSecret, isLetter, getSecret
```

## Expected Output

```
LEVEL 2 UNLOCKED
IPE
```



# LEVEL 3

## Description

Inheritance + multi transformations.

## Code with Blanks

```
class BaseClue{
    String process(String input){
        char[] arr=input.toCharArray();
        for(int i=0;i<arr.length/2;i++){
            char t=arr[i];
            arr[i]=arr[arr.length-1-i];
            arr[arr.length-1-i]=t;
        }
        return new String(arr);
    }
}
class AdvancedClue _____ BaseClue{
    String modify(String input){
        String out="";
        for(int i=0;i<input.length();i++){
            if(i%2==____){
                out+=input.charAt(i);
            }
        }
        return out;
    }
    void reveal(){
        int[] data={89,69,76,76,79,87,32,70,82,85,73,84};
        String text="";
        for(int i=0;i<data.length;i++){
            text+=(char)data[i];
        }
        String s=text.replace(" ","");
        String m=modify(s);
        String f=process(m);
        System.out.println("LEVEL 3 UNLOCKED");
        System.out.println(f);
    }
}
public class Main{
    public static void main(String[] args){
        AdvancedClue obj=new _____();
        obj.reveal();
    }
}
```

## Filled Code

extends, 0, AdvancedClue

## Fill Answers

extends, 0, AdvancedClue

## Expected Output

LEVEL 3 UNLOCKED

TUFOLY

# LEVEL 4

## Description

Polymorphism + traps.

## Code with Blanks

```
class Base{
    String process(String input){
        char[] arr=input.toCharArray();
        for(int i=0;i<arr.length-1;i+=2){
            char t=arr[i];
            arr[i]=arr[i+1];
            arr[i+1]=t;
        }
        return new String(arr);
    }
    void show(){
        System.out.println("WRONG");
    }
}
class RealClue extends Base{
    @_____
    _____ show(){
        int[] data={71,65,77,69,32,80,76,65,89};
        String text="";
        for(int i=0;i<data.length;i++){
            text+=(char)data[i];
        }
        String s=text.replace(" ","");
        String p=process(s);
        String out="";
        for(int i=0;i<p.length();i++){
            if(i%2==____){
                out+=p.charAt(i);
            }
        }
        System.out.println("LEVEL 4 UNLOCKED");
        System.out.println(out);
    }
}
class FakeClue extends Base{
    void show(){
        System.out.println("LEVEL 4 UNLOCKED");
        System.out.println("PLAYGAME");
    }
}
public class Main{
    public static void main(String[] args){
        Base obj=new _____();
        obj.show();
    }
}
```

## Filled Code

Override, void, 1, RealClue

## Fill Answers

Override, void, 1, RealClue

## Expected Output

LEVEL 4 UNLOCKED  
APGM